

1. Show that a finite simple group whose order is at least $r!$ cannot have a proper subgroup of index r .

Solution:

Suppose for contradiction that G is a finite simple group with $|G| \geq r!$ and H is a proper subgroup of G with $[G : H] = r$. Consider the action of G on the set G/H of left cosets of H by left multiplication. This action yields a homomorphism $\varphi : G \rightarrow \mathfrak{S}_r$. Since G acts transitively on G/H (given any two cosets aH and bH , the element ba^{-1} sends aH to bH), the action is nontrivial, so $\ker(\varphi) \neq G$. Since G is simple, the only normal subgroups of G are $\{e\}$ and G itself, so we must have $\ker(\varphi) = \{e\}$. Therefore φ is injective, which means G embeds into $\mathfrak{S}_r$, and in particular $|G| \leq |\mathfrak{S}_r| = r!$. But this contradicts our assumption that $|G| \geq r!$, since equality would force $G \cong \mathfrak{S}_r$, which is not simple for $r \geq 3$ (as $A_r \triangleleft \mathfrak{S}_r$ is a proper nontrivial normal subgroup). For $r = 2$, having a subgroup of index 2 means $H \triangleleft G$, contradicting simplicity since H is proper and nontrivial (as $|G| \geq 2! = 2$ and $|H| = |G|/2 \geq 1$). Hence no such subgroup can exist.

2. Let $n \geq 5$ and assume that A_n is simple. Show that $\mathfrak{S}_n$ has no proper subgroup of index less than n other than A_n . (Hint: use previous question)

Solution:

Suppose H is a proper subgroup of $\mathfrak{S}_n$ with index less than n . Then $[\mathfrak{S}_n : H] < n$ and consider the action of $\mathfrak{S}_n$ on the left cosets of H by left multiplication. This gives us a homomorphism from $\mathfrak{S}_n$ to the symmetric group on the set of cosets, which has order $[\mathfrak{S}_n : H]$. Since the index of H is less than n , this homomorphism cannot be injective (as $\mathfrak{S}_n$ has order $n!$ and the symmetric group on fewer than n elements has order less than $n!$). Therefore, the kernel of this homomorphism is a nontrivial normal subgroup K of $\mathfrak{S}_n$. Since both K and A_n are normal so is its intersection $K \cap A_n$, which is also normal in A_n . The simplicity of A_n implies that $K \cap A_n$ is either trivial or all of A_n . If $K \cap A_n$ is trivial, then K is a subgroup of order 2, which is impossible since $n \geq 5$ and no single permutation of order 2 will form a normal subgroup of $\mathfrak{S}_n$. Therefore, $K \cap A_n = A_n$, which implies $A_n \subseteq K$. Since $[\mathfrak{S}_n : A_n] = 2$, the only subgroups of $\mathfrak{S}_n$ containing A_n are A_n and $\mathfrak{S}_n$ itself. If $K = \mathfrak{S}_n$, then φ is trivial, contradicting the fact that the action on cosets is transitive (hence nontrivial). So $K = A_n$. Now, since $K = \ker(\varphi) = \bigcap_{g \in \mathfrak{S}_n} gHg^{-1} \subseteq H$, we conclude $A_n \subseteq H$. Since H is a proper subgroup and $[\mathfrak{S}_n : A_n] = 2$, this forces $H = A_n$.

Thus, the only proper subgroup of $\mathfrak{S}_n$ with index less than n is A_n .

3. Let $D_4 = \langle r, s \mid r^4 = s^2 = e, srs = r^{-1} \rangle$ be the dihedral group of order 8 (symmetries of the square). Consider D_4 acting on the set of colorings of the 4 vertices of a square using 3 colors.
- For each element $g \in D_4$, compute the number $|\text{Fix}(g)|$ of colorings fixed by g .
 - Use Burnside's lemma to determine the number of distinct colorings of the vertices of a square with 3 colors, up to symmetry.
 - State the general formula for n colors.

Solution:

The elements of D_4 are $\{e, r, r^2, r^3, s, sr, sr^2, sr^3\}$, where r is rotation by 90° and s is a reflection. We label the vertices 1, 2, 3, 4 going clockwise. A coloring is a function $f : \{1, 2, 3, 4\} \rightarrow \{c_1, c_2, c_3\}$, so there are $3^4 = 81$ total colorings.

(a) We compute $|\text{Fix}(g)|$ for each $g \in D_4$:

- e : Every coloring is fixed. $|\text{Fix}(e)| = 3^4 = 81$.
- $r = (1234)$: A coloring is fixed if and only if all four vertices have the same color. $|\text{Fix}(r)| = 3$.
- $r^2 = (13)(24)$: A coloring is fixed if and only if vertices 1, 3 share a color and vertices 2, 4 share a color. $|\text{Fix}(r^2)| = 3^2 = 9$.
- $r^3 = (1432)$: Same reasoning as r . $|\text{Fix}(r^3)| = 3$.
- $s = (24)$ (reflection through the axis through vertices 1 and 3): Vertices 1 and 3 can be any color independently, and vertices 2, 4 must share a color. $|\text{Fix}(s)| = 3^3 = 27$.
- $sr^2 = (13)$ (reflection through the axis through vertices 2 and 4): By symmetry, $|\text{Fix}(sr^2)| = 3^3 = 27$.
- $sr = (12)(34)$ (reflection through a diagonal): Vertices 1, 2 must share a color and vertices 3, 4 must share a color. $|\text{Fix}(sr)| = 3^2 = 9$.
- $sr^3 = (14)(23)$ (reflection through the other diagonal): By symmetry, $|\text{Fix}(sr^3)| = 3^2 = 9$.

(b) By Burnside's lemma, the number of distinct colorings up to symmetry is

$$\frac{1}{|D_4|} \sum_{g \in D_4} |\text{Fix}(g)| = \frac{1}{8} (81 + 3 + 9 + 3 + 27 + 27 + 9 + 9) = \frac{168}{8} = 21.$$

(c) For n colors, the fixed-point counts become $n^4, n, n^2, n, n^3, n^3, n^2, n^2$ respectively. So the number of distinct colorings is

$$\frac{1}{8} (n^4 + 2n^3 + 3n^2 + 2n).$$

One can verify: for $n = 3$ this gives $\frac{1}{8} (81 + 54 + 27 + 6) = \frac{168}{8} = 21$.

4. Find the **distinct** irreducible permutation representations of A_4 . (Take into account that there are 10 different subgroups of A_4 . Find them first)

Solution:

Recall that an irreducible permutation representation (i.e., a transitive A_4 -set) is of the form A_4/H for some subgroup $H \leq A_4$, and two such representations A_4/H and A_4/K are isomorphic if and only if H and K are conjugate in A_4 . So we must first find all subgroups of A_4 , then determine their conjugacy classes.

Step 1: The 10 subgroups of A_4 . Recall $|A_4| = 12$, so by Lagrange's theorem, subgroups can have order 1, 2, 3, 4, 6, or 12.

- *Order 1:* The trivial subgroup $H_1 = \{e\}$.
- *Order 2:* Each is generated by a double transposition. There are exactly three:

$$H_2 = \langle (12)(34) \rangle = \{e, (12)(34)\},$$

$$H_3 = \langle (13)(24) \rangle = \{e, (13)(24)\},$$

$$H_4 = \langle (14)(23) \rangle = \{e, (14)(23)\}.$$

- *Order 3:* Each is generated by a 3-cycle. There are exactly four:

$$H_5 = \langle (123) \rangle = \{e, (123), (132)\},$$

$$H_6 = \langle (124) \rangle = \{e, (124), (142)\},$$

$$H_7 = \langle (134) \rangle = \{e, (134), (143)\},$$

$$H_8 = \langle (234) \rangle = \{e, (234), (243)\}.$$

- *Order 4:* The Klein four-group

$$H_9 = V_4 = \{e, (12)(34), (13)(24), (14)(23)\}.$$

This is the only subgroup of order 4 (it is the unique Sylow 2-subgroup of A_4).

- *Order 6:* There is no subgroup of order 6. Indeed, such a subgroup would have index 2 in A_4 and hence be normal, but one can check that no such subgroup exists (the three double transpositions together with e form V_4 of order 4, and any subgroup of order 6 would need to contain elements of order 2 and 3 in a way that forces it to equal A_4).
- *Order 12:* The whole group $H_{10} = A_4$.

This gives $1 + 3 + 4 + 1 + 0 + 1 = 10$ subgroups in total.

Step 2: The 5 conjugacy classes of subgroups. Two subgroups H and K are conjugate if $gHg^{-1} = K$ for some $g \in A_4$.

1. $\{H_1\} = \{\{e\}\}$: The trivial subgroup is conjugate only to itself.
2. $\{H_2, H_3, H_4\}$: The three subgroups of order 2 are all conjugate. For instance, $(1\ 2\ 3)(1\ 2)(3\ 4)(1\ 3\ 2) = (1\ 3)(2\ 4)$, so $(1\ 2\ 3)H_2(1\ 2\ 3)^{-1} = H_3$.
3. $\{H_5, H_6, H_7, H_8\}$: The four subgroups of order 3 are all conjugate. For instance, $(1\ 2)(3\ 4)\langle(1\ 2\ 3)\rangle(1\ 2)(3\ 4) = \langle(2\ 1\ 4)\rangle = \langle(1\ 2\ 4)\rangle$, so H_5 is conjugate to H_6 .
4. $\{H_9\} = \{V_4\}$: The Klein four-group is normal in A_4 (it is the unique Sylow 2-subgroup), so it is conjugate only to itself.
5. $\{H_{10}\} = \{A_4\}$: The whole group is conjugate only to itself.

Step 3: The 5 distinct irreducible permutation representations. Each conjugacy class yields one isomorphism class of transitive A_4 -set A_4/H :

Representative H	$ H $	$ A_4/H $	Description
$H_1 = \{e\}$	1	12	Regular representation
$H_2 = \langle(1\ 2)(3\ 4)\rangle$	2	6	Action on 6 cosets
$H_5 = \langle(1\ 2\ 3)\rangle$	3	4	Action on 4 cosets
$H_9 = V_4$	4	3	Action on 3 cosets
$H_{10} = A_4$	12	1	Trivial representation

Therefore, there are exactly **5** distinct irreducible permutation representations of A_4 .

5. Let H be a normal subgroup of a finite group G and assume that $|H| = p$, where p is a prime. Show that H is contained in every p -Sylow subgroup of G .

Solution:

Since $|H| = p$, the subgroup H is a p -subgroup of G . By Sylow's theorem, every p -subgroup of G is contained in some Sylow p -subgroup, so there exists a Sylow p -subgroup Q of G with $H \leq Q$. Now let P be any Sylow p -subgroup of G . By Sylow's theorem, all Sylow p -subgroups are conjugate, so $P = gQg^{-1}$ for some $g \in G$. Since H is normal in G , we have $gHg^{-1} = H$, and therefore

$$H = gHg^{-1} \leq gQg^{-1} = P.$$

Since P was an arbitrary Sylow p -subgroup, H is contained in every Sylow p -subgroup of G .

6. Prove that there are not simple groups of order 40.

Solution:

Suppose for contradiction that G is a simple group of order $40 = 2^3 \cdot 5$. By Sylow's theorems, the number n_5 of Sylow 5-subgroups must satisfy:

- $n_5 \equiv 1 \pmod{5}$,
- n_5 divides $|G|/5 = 8$.

The divisors of 8 are 1, 2, 4, 8. Among these, the only one congruent to 1 modulo 5 is $n_5 = 1$ (since $2 \equiv 2$, $4 \equiv 4$, and $8 \equiv 3$ modulo 5). Therefore $n_5 = 1$, meaning there is a unique Sylow 5-subgroup P of G . A unique Sylow subgroup is necessarily normal (since gPg^{-1} is also a Sylow 5-subgroup for every $g \in G$, uniqueness forces $gPg^{-1} = P$). Thus $P \triangleleft G$ with $P \neq \{e\}$ and $P \neq G$, contradicting the simplicity of G . Therefore, there is no simple group of order 40.

7. Show that an abelian group has a composition series if and only if it is finite.

Solution:

($\Rightarrow$) Suppose G is a finite abelian group. We prove by strong induction on $|G|$ that G has a composition series. If $|G| = 1$, the trivial chain $\{e\}$ is a composition series. If $|G| > 1$, let p be a prime dividing $|G|$. By Cauchy's theorem, G contains an element a of order p . Since G is abelian, $\langle a \rangle \triangleleft G$, and the quotient $G/\langle a \rangle$ is an abelian group of order $|G|/p < |G|$. By the induction hypothesis, $G/\langle a \rangle$ has a composition series

$$\{e\} = \overline{G}_0 \triangleleft \overline{G}_1 \triangleleft \cdots \triangleleft \overline{G}_k = G/\langle a \rangle.$$

By the correspondence theorem, each $\overline{G}_i$ corresponds to a subgroup G_i of G containing $\langle a \rangle$, with $G_i/\langle a \rangle = \overline{G}_i$. Moreover, $\langle a \rangle \cong \mathbb{Z}_p$ is simple (as it is cyclic of prime order and abelian). Therefore the chain

$$\{e\} \triangleleft \langle a \rangle = G_0 \triangleleft G_1 \triangleleft \cdots \triangleleft G_k = G$$

is a composition series for G , since each factor $G_{i+1}/G_i \cong \overline{G}_{i+1}/\overline{G}_i$ is simple, and $\langle a \rangle/\{e\} \cong \mathbb{Z}_p$ is simple.

($\Leftarrow$) Suppose G is an abelian group with a composition series

$$\{e\} = G_0 \triangleleft G_1 \triangleleft \cdots \triangleleft G_n = G.$$

Each factor G_{i+1}/G_i is simple. Since G is abelian, every subgroup and quotient is abelian, so each G_{i+1}/G_i is a simple abelian group, hence cyclic of prime order p_i . Therefore $|G| = \prod_{i=0}^{n-1} |G_{i+1}/G_i| = p_0 p_1 \cdots p_{n-1}$, which is a finite product of primes. In particular, G is finite.

8. Find all of the abelian groups of order 720 up to isomorphism.

Solution:

By the fundamental theorem of finite abelian groups, every finite abelian group can be expressed as a direct product of cyclic groups of prime power order. The prime factorization of 720 is $2^4 \cdot 3^2 \cdot 5^1$. We can find all abelian groups of order 720 by considering the possible direct products of cyclic groups corresponding to each prime power. For the prime 2^4 , the possible abelian groups are:

- $\mathbb{Z}_{16}$
- $\mathbb{Z}_8 \times \mathbb{Z}_2$
- $\mathbb{Z}_4 \times \mathbb{Z}_4$
- $\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_2$
- $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$

For the prime 3^2 , the possible abelian groups are:

- $\mathbb{Z}_9$
- $\mathbb{Z}_3 \times \mathbb{Z}_3$

For the prime 5^1 , the only abelian group is $\mathbb{Z}_5$. Combining these, the abelian groups of order 720 up to isomorphism are:

- $\mathbb{Z}_{16} \times \mathbb{Z}_9 \times \mathbb{Z}_5$
- $\mathbb{Z}_{16} \times (\mathbb{Z}_3 \times \mathbb{Z}_3) \times \mathbb{Z}_5$
- $(\mathbb{Z}_8 \times \mathbb{Z}_2) \times \mathbb{Z}_9 \times \mathbb{Z}_5$
- $(\mathbb{Z}_8 \times \mathbb{Z}_2) \times (\mathbb{Z}_3 \times \mathbb{Z}_3) \times \mathbb{Z}_5$
- $(\mathbb{Z}_4 \times \mathbb{Z}_4) \times \mathbb{Z}_9 \times \mathbb{Z}_5$
- $(\mathbb{Z}_4 \times \mathbb{Z}_4) \times (\mathbb{Z}_3 \times \mathbb{Z}_3) \times \mathbb{Z}_5$
- $(\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_2) \times \mathbb{Z}_9 \times \mathbb{Z}_5$
- $(\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_2) \times (\mathbb{Z}_3 \times \mathbb{Z}_3) \times \mathbb{Z}_5$
- $(\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2) \times \mathbb{Z}_9 \times \mathbb{Z}_5$
- $(\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2) \times (\mathbb{Z}_3 \times \mathbb{Z}_3) \times \mathbb{Z}_5$

This gives $5 \times 2 \times 1 = 10$ abelian groups of order 720 up to isomorphism.

9. Let G be an abelian group of order m . If n divides m , prove that G has a subgroup of order n .

Solution:

By the fundamental theorem of finite abelian groups, we can write

$$G \cong \mathbb{Z}_{p_1^{a_{1,1}}} \times \mathbb{Z}_{p_1^{a_{1,2}}} \times \cdots \times \mathbb{Z}_{p_r^{a_{r,s_r}}},$$

where $m = \prod_{i=1}^r p_i^{k_i}$ with $k_i = a_{i,1} + a_{i,2} + \cdots + a_{i,s_i}$ and $a_{i,1} \geq a_{i,2} \geq \cdots \geq a_{i,s_i} \geq 1$.

Since $n \mid m$, we can write $n = p_1^{l_1} p_2^{l_2} \cdots p_r^{l_r}$ where $0 \leq l_i \leq k_i$ for each i . It suffices to show that for each prime p_i , the p_i -primary component $G_{p_i} \cong \mathbb{Z}_{p_i^{a_{i,1}}} \times \cdots \times \mathbb{Z}_{p_i^{a_{i,s_i}}}$ has a subgroup of order $p_i^{l_i}$, since the product of these subgroups (across distinct primes) will be a subgroup of G of order n .

Fix a prime $p = p_i$ and write $G_p \cong \mathbb{Z}_{p^{a_1}} \times \cdots \times \mathbb{Z}_{p^{a_s}}$ with $a_1 + \cdots + a_s = k_i$ and $l = l_i \leq k_i$. We need to find nonneg integers $b_1, \dots, b_s$ with $0 \leq b_j \leq a_j$ for each j and $b_1 + \cdots + b_s = l$. This is always possible: greedily set $b_j = \min(a_j, l - b_1 - \cdots - b_{j-1})$ for $j = 1, \dots, s$. Since each cyclic group $\mathbb{Z}_{p^{a_j}}$ has a unique subgroup of order p^{b_j} (namely $\langle p^{a_j-b_j} \rangle$), the subgroup

$$\langle p^{a_1-b_1} \rangle \times \cdots \times \langle p^{a_s-b_s} \rangle \leq G_p$$

has order $p^{b_1+\cdots+b_s} = p^l$.

Taking the product of such subgroups over all primes p_i gives a subgroup of G of order $\prod_{i=1}^r p_i^{l_i} = n$.

10. Find a composition series for each of the following groups and determine the isomorphism type of each factor group.

(a) $\mathbb{Z}_{12}$

(b) The quaternions, Q_8

(c) D_4

(d) $S_3 \times \mathbb{Z}_4$

(e) S_4

(f) S_n , $n \geq 5$

Solution:

Recall that a composition series is a chain of subgroups $\{e\} = G_0 \triangleleft G_1 \triangleleft \cdots \triangleleft G_n = G$ where each factor G_{i+1}/G_i is simple. For abelian groups, the simple factors are cyclic of prime order.

- (a) $\mathbb{Z}_{12}$: Since $12 = 2^2 \cdot 3$, we need factors $\mathbb{Z}_2, \mathbb{Z}_2, \mathbb{Z}_3$. A composition series is

$$\{0\} \triangleleft \langle 6 \rangle \triangleleft \langle 2 \rangle \triangleleft \mathbb{Z}_{12},$$

where $\langle 6 \rangle = \{0, 6\}$ has order 2 and $\langle 2 \rangle = \{0, 2, 4, 6, 8, 10\}$ has order 6. The factors are:

$$\langle 6 \rangle / \{0\} \cong \mathbb{Z}_2, \quad \langle 2 \rangle / \langle 6 \rangle \cong \mathbb{Z}_3, \quad \mathbb{Z}_{12} / \langle 2 \rangle \cong \mathbb{Z}_2.$$

- (b) The quaternions, Q_8 : Since $|Q_8| = 8 = 2^3$, we need three factors of $\mathbb{Z}_2$. The subgroups $\langle i \rangle, \langle j \rangle, \langle k \rangle$ each have order 4 and all contain $\langle -1 \rangle = \{1, -1\}$, but they are *not* nested. A composition series is

$$\{1\} \triangleleft \langle -1 \rangle \triangleleft \langle i \rangle \triangleleft Q_8,$$

where $\langle -1 \rangle = \{1, -1\}$ has order 2 and $\langle i \rangle = \{1, -1, i, -i\}$ has order 4. The factors are:

$$\langle -1 \rangle / \{1\} \cong \mathbb{Z}_2, \quad \langle i \rangle / \langle -1 \rangle \cong \mathbb{Z}_2, \quad Q_8 / \langle i \rangle \cong \mathbb{Z}_2.$$

(One could equally use $\langle j \rangle$ or $\langle k \rangle$ in place of $\langle i \rangle$.)

- (c) D_4 : Since $|D_4| = 8 = 2^3$, we again need three factors of $\mathbb{Z}_2$. A composition series is

$$\{e\} \triangleleft \langle r^2 \rangle \triangleleft \langle r \rangle \triangleleft D_4,$$

where $\langle r^2 \rangle = \{e, r^2\}$ has order 2 and $\langle r \rangle = \{e, r, r^2, r^3\}$ has order 4. Note that $\langle r \rangle \triangleleft D_4$ since $[D_4 : \langle r \rangle] = 2$, and $\langle r^2 \rangle \triangleleft \langle r \rangle$ since $\langle r \rangle$ is abelian. The factors are:

$$\langle r^2 \rangle / \{e\} \cong \mathbb{Z}_2, \quad \langle r \rangle / \langle r^2 \rangle \cong \mathbb{Z}_2, \quad D_4 / \langle r \rangle \cong \mathbb{Z}_2.$$

- (d) $S_3 \times \mathbb{Z}_4$: Since $|S_3 \times \mathbb{Z}_4| = 24 = 2^3 \cdot 3$, we need factors $\mathbb{Z}_2, \mathbb{Z}_2, \mathbb{Z}_2, \mathbb{Z}_3$. A composition series is

$$\{(e, 0)\} \triangleleft \langle (e, 2) \rangle \triangleleft \{e\} \times \mathbb{Z}_4 \triangleleft A_3 \times \mathbb{Z}_4 \triangleleft S_3 \times \mathbb{Z}_4,$$

where:

- $\langle (e, 2) \rangle = \{(e, 0), (e, 2)\}$ has order 2,
- $\{e\} \times \mathbb{Z}_4 = \{(e, 0), (e, 1), (e, 2), (e, 3)\}$ has order 4,
- $A_3 \times \mathbb{Z}_4$ has order 12 (here $A_3 = \langle (1\ 2\ 3) \rangle \cong \mathbb{Z}_3$).

The factors are:

$$\mathbb{Z}_2, \quad \mathbb{Z}_2, \quad \mathbb{Z}_3, \quad \mathbb{Z}_2.$$

- (e) S_4 : Since $|S_4| = 24 = 2^3 \cdot 3$ and S_4 is not simple, we must refine further. Let $V_4 = \{e, (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\} \triangleleft A_4 \triangleleft S_4$. Since $V_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, we can refine it further. A composition series is

11. Let X be a set. Define

$$\mathbb{C}X = \{\alpha_1 x_1 + \cdots + \alpha_k x_k \mid \alpha_i \in \mathbb{C}, x_i \in X \forall i = 1, \dots, k, k \geq 0\}.$$

Here the elements of X are treated as formal symbols, and the expressions above are formal $\mathbb{C}$ -linear combinations (i.e., two such expressions are equal if and only if they assign the same coefficient to each element of X , where omitted elements are understood to have coefficient 0). Show that $\mathbb{C}X$ has a natural structure of a vector space over $\mathbb{C}$.

Solution:

We regard an element of $\mathbb{C}X$ as a function $f : X \rightarrow \mathbb{C}$ with finite support, where the formal linear combination $\alpha_1 x_1 + \cdots + \alpha_k x_k$ corresponds to the function f defined by $f(x_i) = \alpha_i$ for $i = 1, \dots, k$ and $f(x) = 0$ for all $x \notin \{x_1, \dots, x_k\}$. Under this identification, we define vector addition and scalar multiplication pointwise.

Addition. Given $f, g \in \mathbb{C}X$, define $(f + g) : X \rightarrow \mathbb{C}$ by

$$(f + g)(x) = f(x) + g(x) \quad \text{for all } x \in X.$$

Since f and g each have finite support, the function $f + g$ also has finite support (it is supported on $\text{supp}(f) \cup \text{supp}(g)$, which is a finite union of finite sets), so $f + g \in \mathbb{C}X$.

In terms of formal sums: if $f = \sum_{x \in X} \alpha_x x$ and $g = \sum_{x \in X} \beta_x x$ (where all but finitely many coefficients are zero), then

$$f + g = \sum_{x \in X} (\alpha_x + \beta_x) x.$$

Scalar multiplication. Given $\lambda \in \mathbb{C}$ and $f \in \mathbb{C}X$, define $(\lambda f) : X \rightarrow \mathbb{C}$ by

$$(\lambda f)(x) = \lambda \cdot f(x) \quad \text{for all } x \in X.$$

Since $\text{supp}(\lambda f) \subseteq \text{supp}(f)$, the function λf has finite support, so $\lambda f \in \mathbb{C}X$.

Verification of the vector space axioms. We check all eight axioms.

1. *Associativity of addition:* For all $f, g, h \in \mathbb{C}X$ and $x \in X$,

$$((f + g) + h)(x) = (f(x) + g(x)) + h(x) = f(x) + (g(x) + h(x)) = (f + (g + h))(x),$$

using associativity of addition in $\mathbb{C}$.

2. *Commutativity of addition:* $(f + g)(x) = f(x) + g(x) = g(x) + f(x) = (g + f)(x)$, using commutativity in $\mathbb{C}$.

3. *Additive identity:* The zero vector is the function $\mathbf{0} : X \rightarrow \mathbb{C}$ defined by $\mathbf{0}(x) = 0$ for all $x \in X$ (the empty formal sum). Then $(f + \mathbf{0})(x) = f(x) + 0 = f(x)$ for all x , so $f + \mathbf{0} = f$.

4. *Additive inverses:* Given $f \in \mathbb{C}X$, define $(-f)(x) = -f(x)$. Then $(f + (-f))(x) = f(x) - f(x) = 0$ for all x , so $f + (-f) = \mathbf{0}$. Since $-f$ has the same (finite) support as f , we have $-f \in \mathbb{C}X$.

5. *Compatibility of scalar multiplication:* For $\lambda, \mu \in \mathbb{C}$ and $f \in \mathbb{C}X$,

$$(\lambda(\mu f))(x) = \lambda \cdot (\mu f)(x) = \lambda \cdot (\mu \cdot f(x)) = (\lambda\mu) \cdot f(x) = ((\lambda\mu)f)(x),$$

using associativity of multiplication in $\mathbb{C}$.

6. *Identity element of scalar multiplication:* $(1 \cdot f)(x) = 1 \cdot f(x) = f(x)$ for all x , so $1 \cdot f = f$.

7. *Distributivity of scalar multiplication over vector addition:* For $\lambda \in \mathbb{C}$ and $f, g \in \mathbb{C}X$,

$$(\lambda(f + g))(x) = \lambda(f(x) + g(x)) = \lambda f(x) + \lambda g(x) = (\lambda f + \lambda g)(x),$$

using distributivity in $\mathbb{C}$.

8. *Distributivity of scalar multiplication over field addition:* For $\lambda, \mu \in \mathbb{C}$ and $f \in \mathbb{C}X$,

$$((\lambda + \mu)f)(x) = (\lambda + \mu)f(x) = \lambda f(x) + \mu f(x) = (\lambda f + \mu f)(x),$$

again using distributivity in $\mathbb{C}$.

Since all eight axioms are satisfied, $\mathbb{C}X$ is a vector space over $\mathbb{C}$. Note that X itself naturally forms a basis for this space: the elements of X are linearly independent by definition (a formal sum $\alpha_1 x_1 + \cdots + \alpha_k x_k = \mathbf{0}$ implies $\alpha_i = 0$ for all i), and every element of $\mathbb{C}X$ is by construction a finite linear combination of elements of X .